Statistical Signal Processing
Lecture 03:
Random signals in frequency-domain: power spectral density + filtering of random signals

Literature and Problems: See course schedule

Agenda (preliminary):

Power spectral density $S_{xx}(\omega)$

- motivation
- definitions: general, continuous-time and discrete-time
- units
- explanation 1 for definition: Wiener-Khinchine theorem
- explanation 2 for definition: total signal power = $R_{xx}(0) = \int S_{xx}(f)\, df$
- calculation of signal power in limited bandwidth
- explanation 3 for definition: correlation versus convolution
- examples/demos
- general properties for $S_{xx}(\omega)$
- MATLAB calculation of $S_{xx}(\omega)$: *freqz, fft, pwelch*

Cross spectral density $S_{xy}(\omega)$

- definition
- general properties for $S_{xy}(\omega)$

Bandwidth of random signal

- LP signals and BP signals
- bandwidth for ideal signals
- bandwidth for practical signals: 3dB BW, RMS BW, Noise equivalent (effective) BW

Filtering of random signals *filter*

- problem setup
- general IO-results: μ_y, $R_{xy}(\tau)$, $R_{yx}(\tau)$, $R_{yy}(\tau)$, σ_y^2, $S_{yy}(\omega)$, $S_{xy}(\omega)$, P_y
- special case: X(t) is white noise (Gaussian or other)
- bandwidth of filter: noise equivalent (effective) BW

Coherence function γ_{xy}^2 *mscohere*

- intuitive explanation
- pure filtering (no noise on input or output signal measurements): $\gamma_{xy}^2 = 1$
- γ_{xy}^2 always between 0 and 1
- $\gamma_{xy}^2 < 1$ if noise on signal measurements or filtering is non-linear